Diabetes MythBuster: Automated Fact-Checking for Arabic Videos About Diabetes

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Abstract

We present an automated fact-checking platform for short Arabic videos about diabetes. The system integrates **Whisper-Large** for transcription, **Gemini Flash 1.5** for claim extraction, and **hybrid retrieval combining real-time web-search from trusted sources and RAG from scientific literature**. Experiments show Whisper-Large and Gemini Flash 1.5 combined with retrieval as effective, enabling reliable claim verification with medical reasoning.

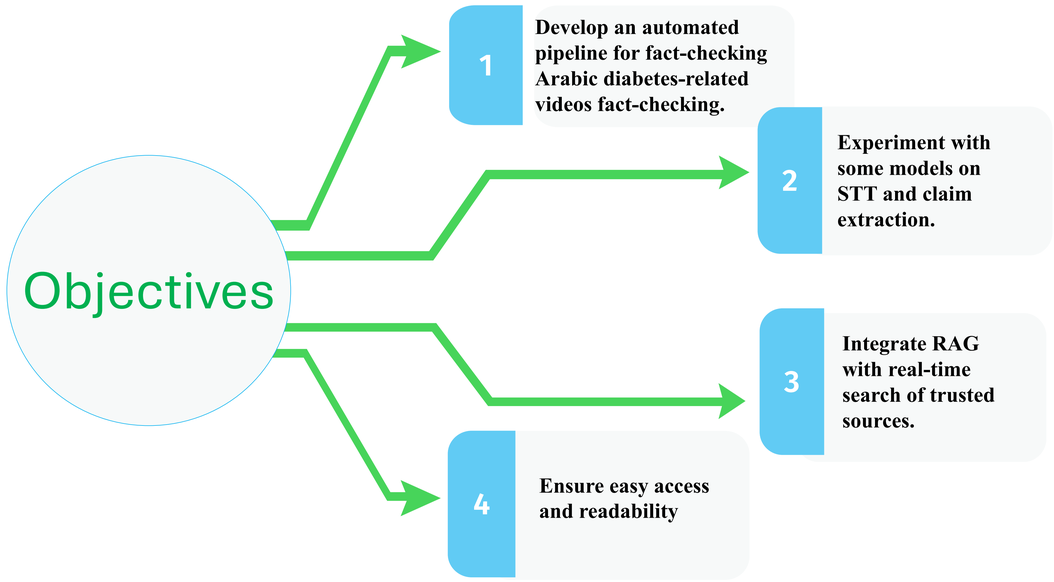
Problem

- Rapid spread of **misinformation** in Arabic diabetes-related videos, for some people. **causing fatal consequences**
- Lack of tools to verify short video content automatically.
- Most of the existing fact-checking systems don't focus on trusted medical resources and are mostly for English.

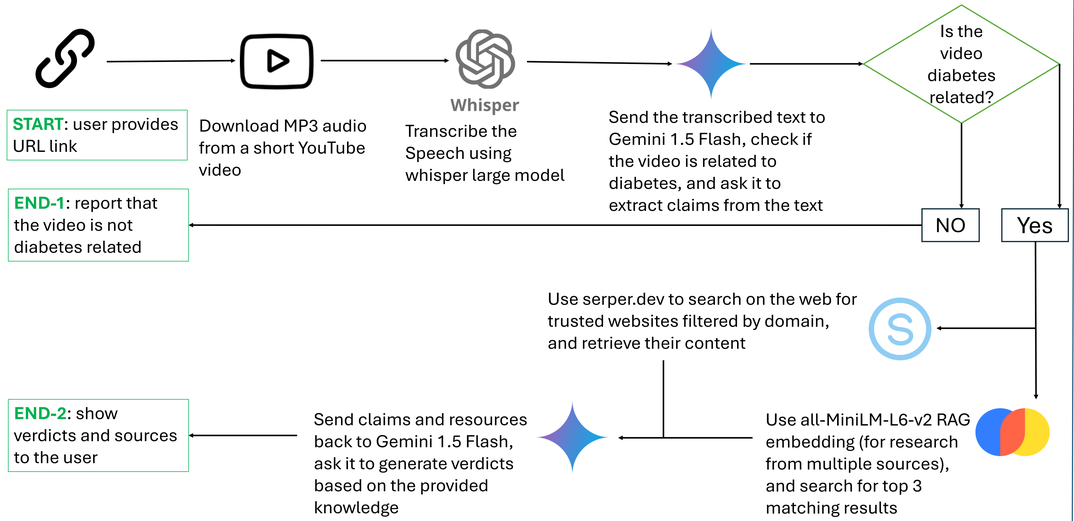
Solution

Accurate STT using Whisper-Large for Arabic videos.
Claim Extraction via Gemini Flash 1.5 with JSON-structured outputs.
Fact-Checking through hybrid retrieval:.
Real-time trusted web search via **Serper.dev.**
RAG, by ChromaDB research corpus with **all-MiniLM-L6-v2** embeddings on **arXiv** literature.
Web User Interface: Interactive frontend and backend pipeline providing real-time fact verification.

Objectives



How our platform works

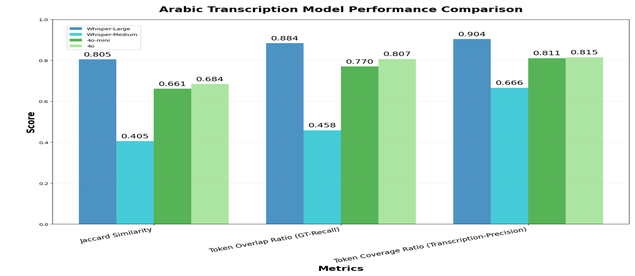


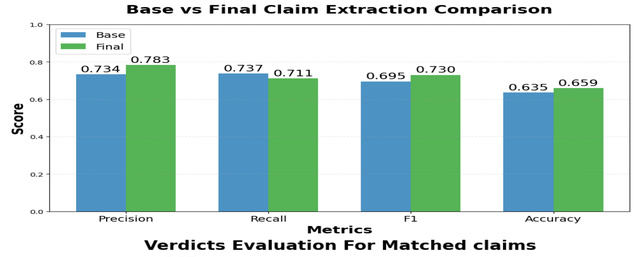
Tools

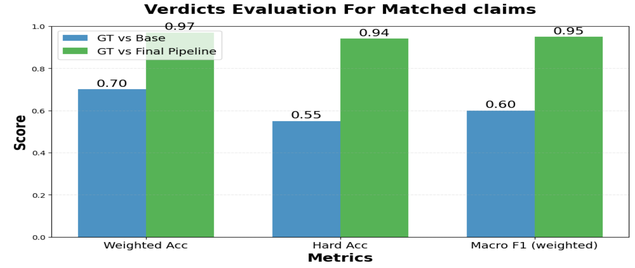
docker
PostgreSQL

References

Results







Prototype

